MA 5104: Statistical Inference and Learning

Module 1: Statistical Models and Likelihood
Lectures 3-5: Understanding Statistical Models

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Week 1 Summary

Human intuition can be misleading

↓

The world contains uncertainty and variability

↓

Probability provides a language for uncertainty

↓

Probability models describe random phenomena

↓

Next Question: What exactly is a statistical model?

Key Takeaways

By the end of this lecture, you should be able to

$\rightarrow$ explain what a statistical model is and why statisticians use models;

$\rightarrow$ interpret the mathematical formulation of a statistical model;

$\rightarrow$ distinguish observations, random variables, parameters, and modelling assumptions;

$\rightarrow$ recognize different classes of statistical models, including parametric, semiparametric, nonparametric, generative, and latent variable models;

$\rightarrow$ appreciate that statistical models are simplified mathematical representations of reality.

Example 1: WYSIATI (What You See Is All There Is)

A company reports

- $\rightarrow$ Revenue growth: 40% per year
- $\rightarrow$ Customer base doubled last year
- $\rightarrow$ Founder graduated from MIT
- $\rightarrow$ Recently secured Series B funding

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- → Strong founder
- → Investor confidence

What We Don't See

- → Profitability, Debt, Cash flow
- → Competition, Data quality
- → Survival probability

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WYSIATI: People construct coherent stories from available information and systematically ignore information that is missing.

Statistics Makes Assumptions Explicit

Human reasoning often relies on implicit assumptions.

Statistics requires every important assumption to be stated explicitly.

Informal Reasoning

Observed facts

$\Downarrow$

Mental story

$\Downarrow$

Decision

Statistical Reasoning

Observed data

$\Downarrow$

Explicit assumptions

$\Downarrow$

Probability model

$\Downarrow$

Inference

Central Question

How do statisticians represent assumptions mathematically?

Representing Assumptions Mathematically

Suppose we observe the systolic blood pressures of n individuals,

$$X_1, \ldots, X_n.$$

A statistician proposes the following mathematical description of the data-generating process:

$$\boxed{X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2).}$$

Question

What information is encoded in this single mathematical statement?

Decoding the Model

$$X_1, \ldots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

Notation	Interpretation
$X_1, \ldots, X_n$	Random variables representing future observations.
iid	Independent observations from the same (identical) population.
$N(\mu, \sigma^2)$	Probability model for each observation.
μ, σ^2	Unknown population parameters.
n	Sample size.

One equation simultaneously specifies the observations, probability model, assumptions, and unknown parameters.

Random Variables, Data and Parameters

For $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$,

Random Variables

$$X_1, \ldots, X_n$$

Before sampling:

- $\rightarrow$ Unknown
- $\rightarrow$ Random
- $\rightarrow$ Described by the model

Parameters

$$\mu, \sigma^2$$

- $\rightarrow$ Unknown
- $\rightarrow$ Fixed constants (frequentist)
- $\rightarrow$ Describe the population
- $\rightarrow$ Do not change after sampling

After observing data, $X_i \rightarrow x_i$.

The values that we calculate from the collected x_i's are merely the estimates for the values of the parameters.

From One Model to Statistical Models

The previous model

$$X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$$

is only one possible statistical model.

Definition

A statistical model is a family of probability distributions

$$\boxed{\mathcal{P} = \{P_\theta : \theta \in \Theta\},}$$

indexed by an unknown parameter θ.

Interpretation

Each value of θ corresponds to one probability distribution. The data are assumed to arise from one (unknown) member of this family.

Probability Models versus Statistical Models

Probability Model	**Statistical Model**
Distribution completely specified	Contains unknown parameters
Used to describe randomness	Used to learn from data
Example $$X \sim N(0, 1)$$	Example $$X \sim N(\mu, \sigma^2)$$
No inference required	Inference required

Key Difference

Statistics begins when the probability model contains **unknown quantities that must be inferred from data**.

Examples of Statistical Models

Blood pressure

$$X_i \sim N(\mu, \sigma^2)$$

Coin tossing

$$X_i \sim Bernoulli(p)$$

Customer arrivals

$$X_i \sim Poisson(\lambda)$$

Linear regression

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Logistic regression

$$P(Y_i = 1) = \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}}$$

Although the mathematical forms differ, every statistical model consists of probability assumptions indexed by unknown parameters.

Why do we need different kinds of models?

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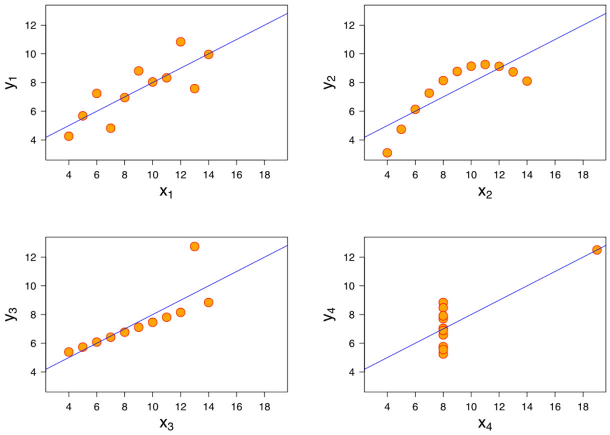
Property	Value	Accuracy
Mean of *x*	9	exact
Sample variance of *x*: s_x^2	11	exact
Mean of *y*	7.50	to 2 decimal places
Sample variance of *y*: s_y^2	4.125	±0.003
Correlation between *x* and *y*	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively
Coefficient of determination of the linear regression: R^2	0.67	to 2 decimal places

All four sets have almost identical statistical parameters. Should we use the same statistical model for these four datasets? Why?

Source: Wikipedia

Anscombe's Quartet

The graphs show them to be considerably different. The same model would almost certainly be problematic.



Lessons from Anscombe's Quartet

Although the four datasets have nearly identical summary statistics,

$$\bar{x}, \quad \bar{y}, \quad s_x^2, \quad s_y^2, \quad r, \quad \hat{y} = \beta_0 + \beta_1 x,$$

their structures are fundamentally different.

Statistical Thinking

→ Numerical summaries never tell the whole story.

→ Every model rests on assumptions.

→ Model choice should be guided by both data and scientific understanding.

What Do We Do with a Statistical Model?

Suppose we wish to learn about the average height of all (IITM) campus residents, $\mu =$ Average height of the population. How should we go about it?

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Based on a sample, which should we report?

- $\rightarrow$ A guess (e.g., 5'3")
- $\rightarrow$ Sample mean
- $\rightarrow$ Sample median/mode
- $\rightarrow$ Confidence interval

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Problem 2: Hypothesis Testing

Suppose someone claims

$$\mu = 5'3".$$

Can the data provide evidence that $\mu \neq 5'3"$, or $\mu > 5'3"$,, or $\mu < 5'3"$?

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Which estimator should we use? How does sample size matter? What makes one estimator better than another? How should we test thes hypotheses?

Classes of Statistical Models

Recall that a statistical model is a family of probability distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\},$$

where θ denotes the unknown quantity (or quantities).

Parametric Model

A statistical model is **parametric** if the unknown parameter is finite-dimensional, $\theta \in \mathbb{R}^p$, where p is fixed.

Example: Blood Pressure

Assume $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, where $\theta = (\mu, \sigma^2)$.

Only two unknown parameters describe the entire population.

Beyond Parametric Models

Semiparametric Model

Some components are finite-dimensional, while others are left unspecified.

$$X_i \sim F, \qquad E(X_i) = \mu,$$

where μ is the parameter of interest but the distribution F is otherwise unknown.

Nonparametric Model

No finite-dimensional parametric form is assumed.
The entire distribution is unknown:

$$X_i \sim F,$$

where F is arbitrary.

Generative Models

A **generative model** specifies the joint distribution

$$P(X, Y), \quad \text{or equivalently} \quad P(X, Y) = P(Y)P(X \mid Y).$$

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What can a generative model do?

- $\rightarrow$ Generate new observations.
- $\rightarrow$ Compute conditional probabilities,
- $\rightarrow$ Simulate from the underlying data-generating process.

Examples: Gaussian Mixture Models, Large Language Models (LLMs)

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Why are LLMs "Generative"?

An LLM models $P(w_t \mid w_1, \ldots, w_{t-1})$, the probability of the next token, allowing it to generate entirely new text.

Pseudo-Random Number Generation

Computers generate observations using a deterministic recursive algorithm.

$$x_{n+1} = g(x_n),$$

starting from an initial value called the **seed**.

$$\boxed{\text{Seed} \to x_1 \to x_2 \to x_3 \to \cdots}$$

A good pseudo-random number generator produces sequences that

- → closely mimic independent random samples;
- → have approximately the desired probability distribution;
- → possess extremely long periods before repeating;
- → are exactly reproducible from the same seed.

From Uniform to Normal Random Variables

Modern computers first generate pseudo-random numbers from

$$U \sim \mathrm{Uniform}(0,1).$$

These are then transformed into samples from other distributions.

$$U(0,1) \quad \Longrightarrow \quad \text{Transformation} \quad \Longrightarrow \quad N(\mu,\sigma^2)$$

One classical method is the **Box–Muller transformation**.
If $U_1, U_2 \stackrel{\text{iid}}{\sim} U(0,1)$, then

$$Z_1 = \sqrt{-2\log U_1}\,\cos(2\pi U_2), \quad Z_2 = \sqrt{-2\log U_1}\,\sin(2\pi U_2),$$

are independent and $Z_1, Z_2 \sim N(0,1)$.

Key Message

Many random number generators begin with Uniform$(0,1)$ variables and obtain other distributions through suitable mathematical transformations.

Pseudo-Random Numbers in R

```
set.seed(123)
```

fixes the initial state of the pseudo-random number generator.

For example, `rnorm(5)` may produce

$$-0.560, \; -0.230, \; 1.559, \; 0.071, \; 0.129.$$

Running `set.seed(123); rnorm(5)` again generates exactly the same sequence.

Why Is This Important?

Pseudo-random numbers make statistical analyses

$\rightarrow$ reproducible, verifiable, and scientifically transparent.

Discriminative Models

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What can a discriminative model do?

- $\rightarrow$ Predict labels or responses.
- $\rightarrow$ Learn decision boundaries.
- $\rightarrow$ Focus on predictive accuracy rather than modelling the data-generating process.

Examples: Logistic Regression, Support Vector Machines, Random Forests, Deep Neural Networks

Key Difference: Marginal distributions assuming X to be given vs. a joint distribution or conditional distribution.

Latent Variable Models

Not every variable in a statistical model is observable. Many models introduce hidden (latent) variables to explain the dependence among the observed variables.

Observed	Latent Variable
Exam scores	

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Observed	Latent Variable
Exam scores	Student ability
Words	Underlying topic
Stock prices	Market state
Gene expression	Biological pathway
Pixel intensities	Object identity

From Models to Data

A statistical model specifies a family of distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

The data

$$x_1, \ldots, x_n$$

have now been observed.

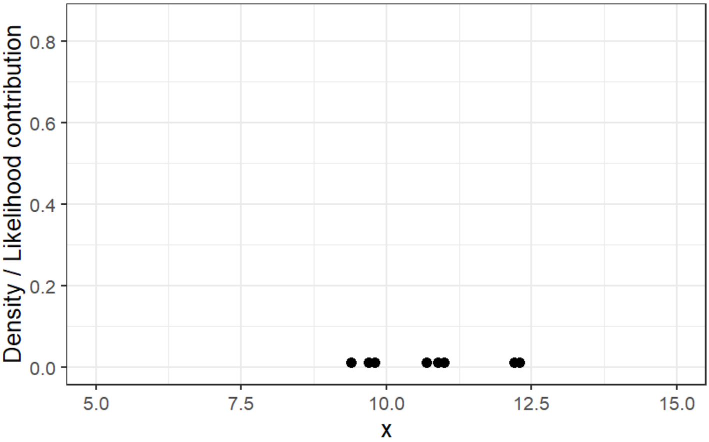
The next question is

$$\boxed{\text{Which parameter values are most plausible given the observed data?}}$$

The answer is provided by the

$$\boxed{\textbf{Likelihood Function}.}$$

Motivating the next question via an example



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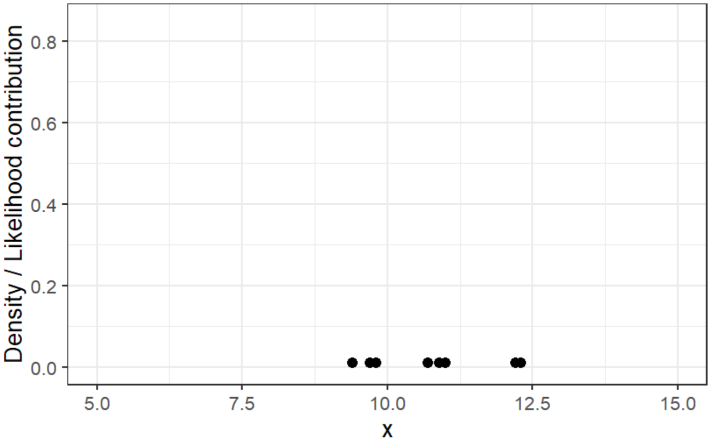

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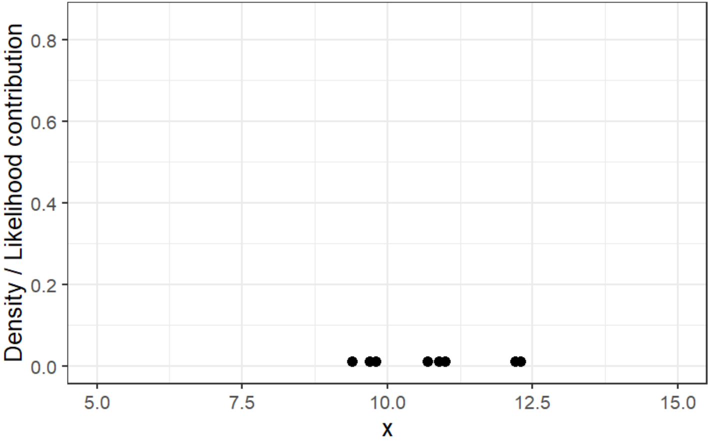

Which probability model generated this data?

Assume that $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, now what can you tell me?

Motivating the next question via an example (contd.)

Which parameter values are most strongly supported by the observed data?

Motivating the next question via an example (contd.)

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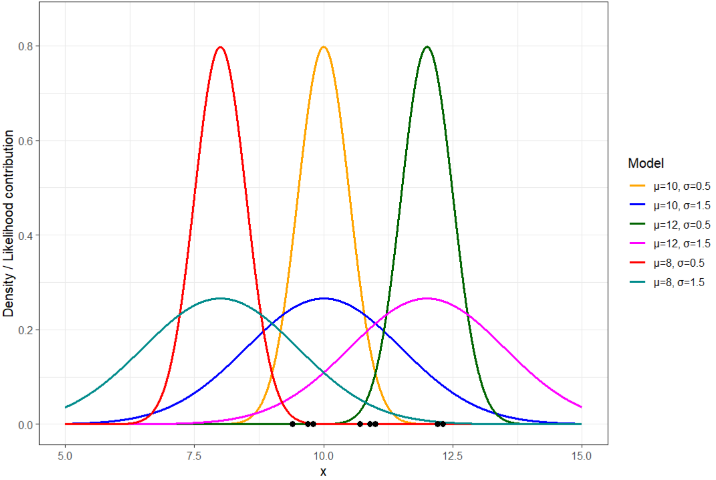
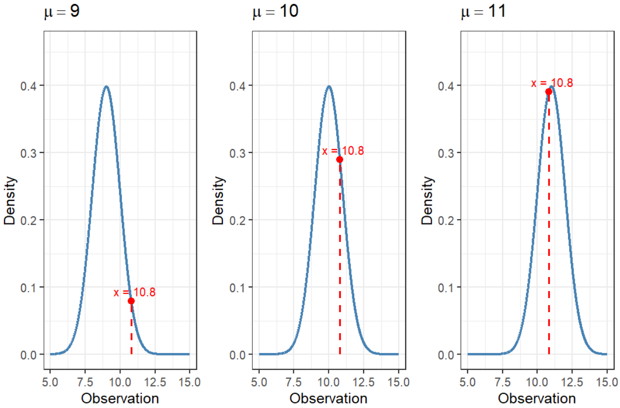


We will discuss parameter estimation in the next module.

The Likelihood Function – Example

Suppose $X \sim N(\mu, 1)$, and we observe $x = 10.8$. Consider $\mu = 9, 10,$ or 11.



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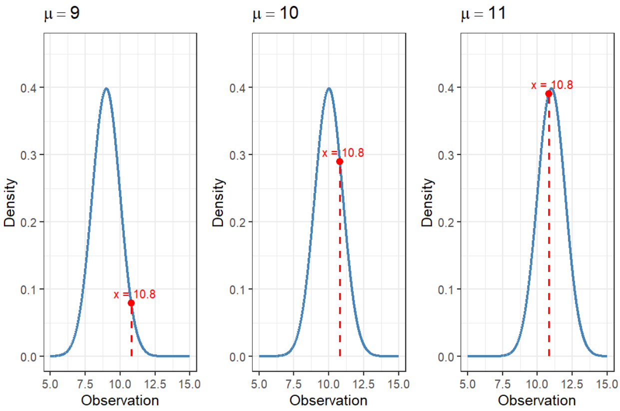

The peak corresponds to the parameter value receiving the greatest support from the observed data.

From Intuition to Mathematics

Our intuition suggests that

$$\mu = 11$$

provides the best explanation for the observation.

Why? Because the observed value $x = 10.8$ lies in a region where the density is highest.

Equivalently,

$$f(10.8; 11, 1) > f(10.8; 10, 1) > f(10.8; 9, 1).$$

Key Idea

A statistical model receives greater support when it assigns a larger density to the observed data.

The Likelihood Function

The observed data are now fixed.

Instead of varying the observation, we compare different values of the parameter.

Definition

For observed data x,

$$\boxed{L(\theta \mid x) = f(x;\theta).}$$

Likelihood uses exactly the same mathematical expression as probability,

$$f(x;\theta),$$

but answers a different question.

Likelihood Is Not a Probability

Probability	Likelihood
Parameter fixed	Data fixed
Observation varies	Parameter varies
Predict future observations	Evaluate competing models
Integrates to one over data	Need not integrate to one over parameters

$\boxed{\text{Likelihood is a measure of relative evidence, not probability.}}$

From One Observation to Many

Suppose the observations are $x_1, x_2, \ldots, x_n$.
Assuming independence,

$$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} f(x_i; \theta).$$

Therefore,

$$\boxed{L(\theta) = \prod_{i=1}^{n} f(x_i; \theta).}$$

Interpretation

Each observation contributes evidence. The likelihood combines evidence from the entire sample.

The Likelihood Principle

A statistical model summarizes uncertainty.

The observed data determine the likelihood.

Likelihood Principle

All evidence in the observed data about the unknown parameter is contained in the likelihood function.

$$L(\theta \mid x).$$

Key Message

Likelihood is the mathematical representation of statistical evidence.

Statistical Demonstration: Fisher's Lady Tasting Tea

Muriel Bristol claimed that she could distinguish between

- $\rightarrow$ tea poured before milk, and
- $\rightarrow$ milk poured before tea,

simply by tasting the tea.

Ronald Fisher proposed a controlled experiment.

`https://jyyna.co.uk/lady-tasting-tea/`

Tea-Tasting Distribution Assuming the Null Hypothesis

Success count	Combinations of selection	Number of Combinations
0	oooo	$1 \times 1 = 1$

Deep-dive

Provides a subject with eight randomly ordered cups of tea – four prepared by pouring milk and then tea, four by pouring tea and then milk.

The subject attempts to select the four cups prepared by one method or the other, and may compare cups directly against each other as desired. Total choices: $\binom{8}{4} = 70$

Tea-Tasting Distribution Assuming the Null Hypothesis

Success count	Combinations of selection	Number of Combinations
0	oooo	$1 \times 1 = 1$ $\binom{4}{0} \times \binom{4}{4}$
1	ooox, ooxo, oxoo, xooo	$4 \times 4 = 16$ $\binom{4}{1} \times \binom{4}{3}$

Why Is This a Statistical Problem?

The experiment involves two competing statistical models.

Model 1

The lady is guessing.

Observed outcomes arise purely due to chance.

Model 2

The lady truly has the claimed ability.

Correct identifications occur more often than expected by chance.

Why Is This a Statistical Problem?

The experiment involves two competing statistical models.

Model 1

The lady is guessing.

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Model 2

The lady truly has the claimed ability.

Correct identifications occur more often than expected by chance.

The observed data provide evidence. The likelihood tells us which model is better supported.

Looking Ahead

In Module 3, we will learn how to decide whether the observed evidence is strong enough to reject the "guessing" model.

Example: Regression to the Mean

Israeli Air Force instructors observed that

- $\rightarrow$ Pilots who performed exceptionally well were often worse on the next flight.
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"Punishment improves performance, whereas praise makes pilots complacent."

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→ Pilots who performed poorly often improved on the next flight.

The instructors concluded:

> *"Punishment improves performance, whereas praise makes pilots complacent."*

Question

Is this necessarily a causal effect, or could it arise purely from randomness?

Why Does Regression to the Mean Occur?

Suppose

$$\text{Observed Performance} = \text{True Ability} + \text{Random Variation.}$$

An exceptionally good performance usually contains

$\rightarrow$ high ability, and

$\rightarrow$ favourable random variation.

Similarly, an exceptionally poor performance often contains

$\rightarrow$ low ability, and

$\rightarrow$ unfavourable random variation.

Statistical Insight

Extreme observations tend to be followed by observations closer to the population average.

All Models Are Wrong

"All models are wrong, but some are useful."

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Every statistical model is an approximation.
No model captures every feature of reality.

Question

If every model is wrong, what makes a model useful?

What Makes a Statistical Model Useful?

Criterion	Why It Matters
Scientific plausibility	Consistent with domain knowledge
Interpretability	Provides meaningful explanations
Parsimony	Avoids unnecessary complexity
Predictive ability	Produces reliable future predictions
Robustness	Stable under small assumption violations
Computational feasibility	Can be fitted efficiently

Key Message

There is rarely a single "best" model. Model choice depends on the scientific objective.

Statistical Thinking versus Algorithmic Thinking

Statistical Thinking	**Algorithmic Thinking**
Probability models	Flexible function classes
Explicit assumptions	Minimal assumptions
Interpretation	Prediction
Uncertainty quantification	Predictive accuracy
Scientific understanding	Automated decision making
Model-driven	Data-driven

Modern Perspective

Machine learning and statistics are complementary. Modern machine learning increasingly incorporates probabilistic modelling, uncertainty quantification, and statistical principles.

Module 1: The Story So Far

Human intuition is often unreliable.

$\Downarrow$

Uncertainty is inherent in real-world phenomena.

$\Downarrow$

Probability provides a language for representing uncertainty.

$\Downarrow$

Statistical models formalize assumptions about the data-generating process.

$\Downarrow$

Likelihood quantifies how well competing models explain the observed data.

Looking Ahead to Module 2

We now know how to represent uncertainty and measure statistical evidence. The next question is:

$\boxed{\text{How do we estimate the unknown parameters?}}$